

AMINES

Amines can be considered as derivatives of ammonia, obtained by replacement of one, two or all the three hydrogen atoms by alkyl and/or aryl groups.

Ex:- $\text{CH}_3\text{-NH}_2$, $\text{C}_6\text{H}_5\text{-NH}_2$, $\text{CH}_3\text{-NH-CH}_3$, $\text{CH}_3\text{-N}\begin{matrix} \text{CH}_3 \\ \text{CH}_3 \end{matrix}$

Structure of Amines:- Nitrogen orbitals in amines are therefore, sp^3 hybridised and the geometry of amines is pyramidal.
Bond angle is 106° in case of trimethylamine.

CLASSIFICATION OF AMINES:-

NH_3
Ammonia

R-NH_2
 1° Amine

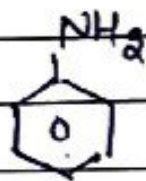
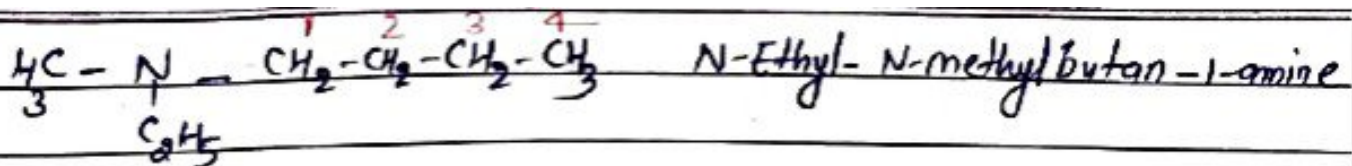
R-NH-R
 2° Amine

R-N-R
 R
 3° Amine

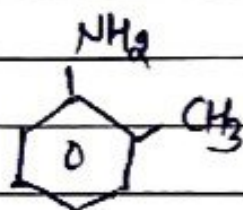
NOMENCLATURE:-

$\text{CH}_3\text{-CH}_2\text{-NH}_2$ Ethanamine

$\text{H}_2\text{N}-\overset{1}{\text{CH}_2}-\overset{2}{\text{CH}}=\overset{3}{\text{CH}_2}$ Prop-2-en-1-amine



Aniline / Benzenamine

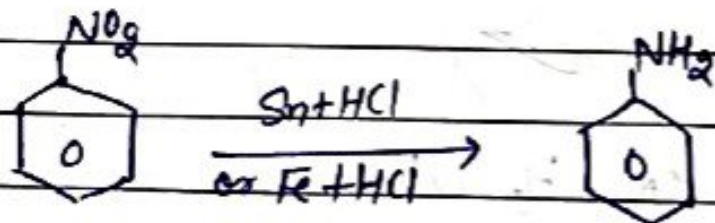
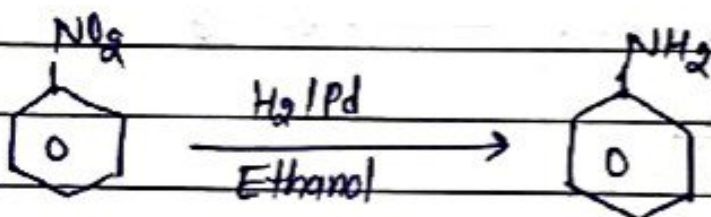


Common Name
o-Toluidine

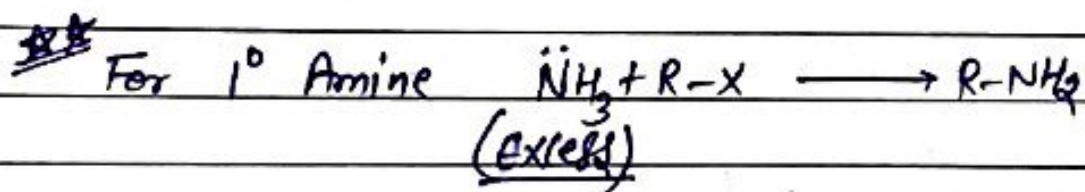
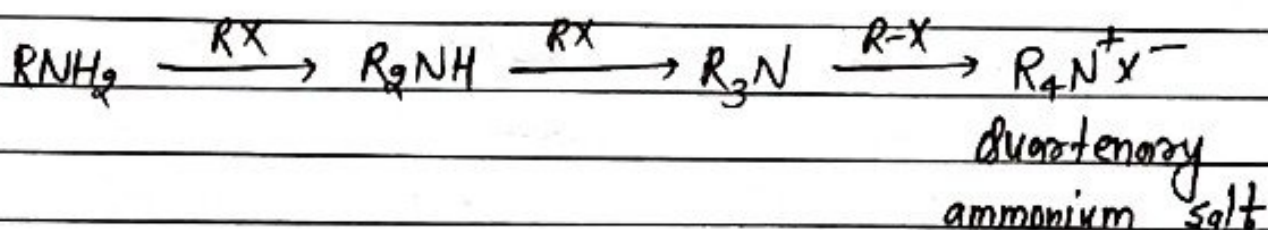
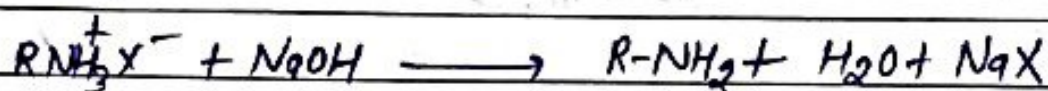
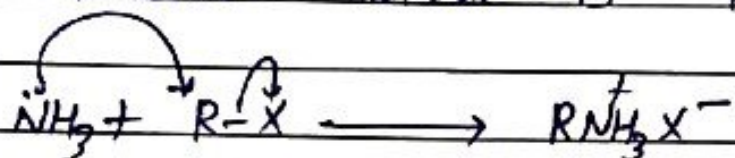
IUPAC
2-Aminotoluene

PREPARATION

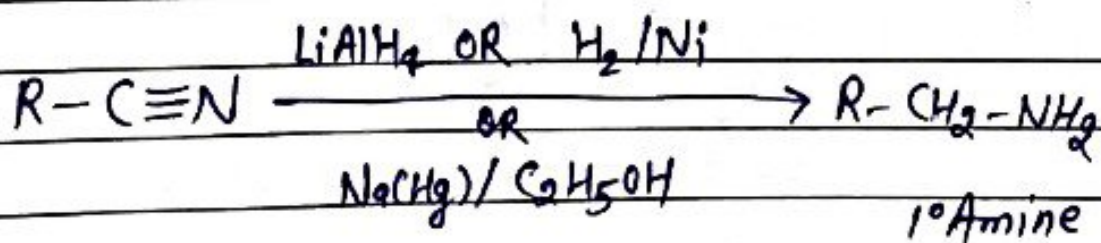
1. REDUCTION OF NITRO COMPOUNDS:-



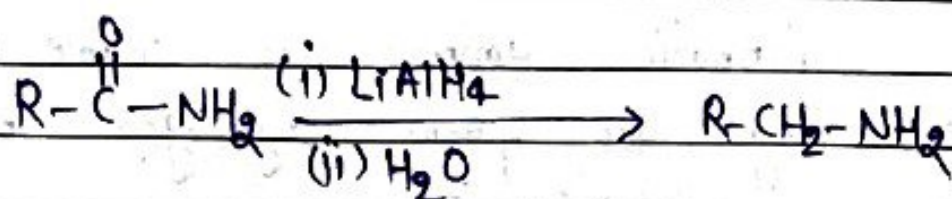
2. Ammonolysis of alkyl Group:- An alkyl or aryl halide on reaction with an ethanolic solution of ammonia undergoes nucleophilic substitution reaction in which the halogen atom is replaced by an amino ($-NH_2$) group. This process of cleavage of the C-X bond by ammonia molecule is known as ammonolysis.



3. Reduction of Nitriles:-

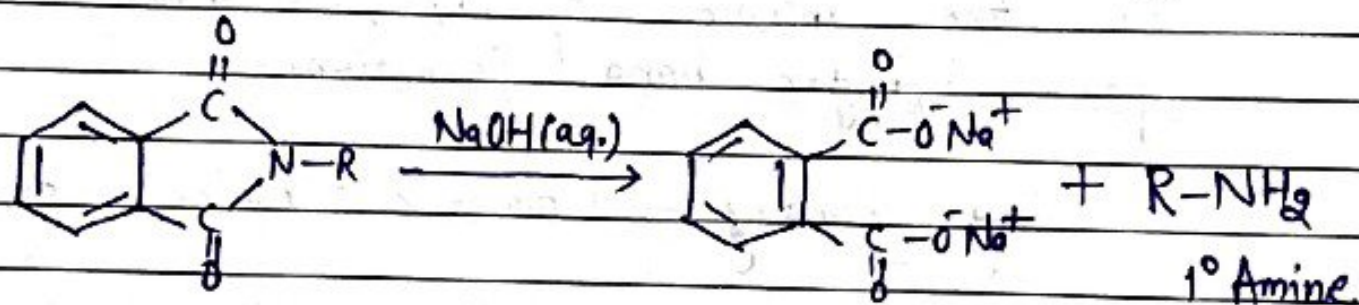
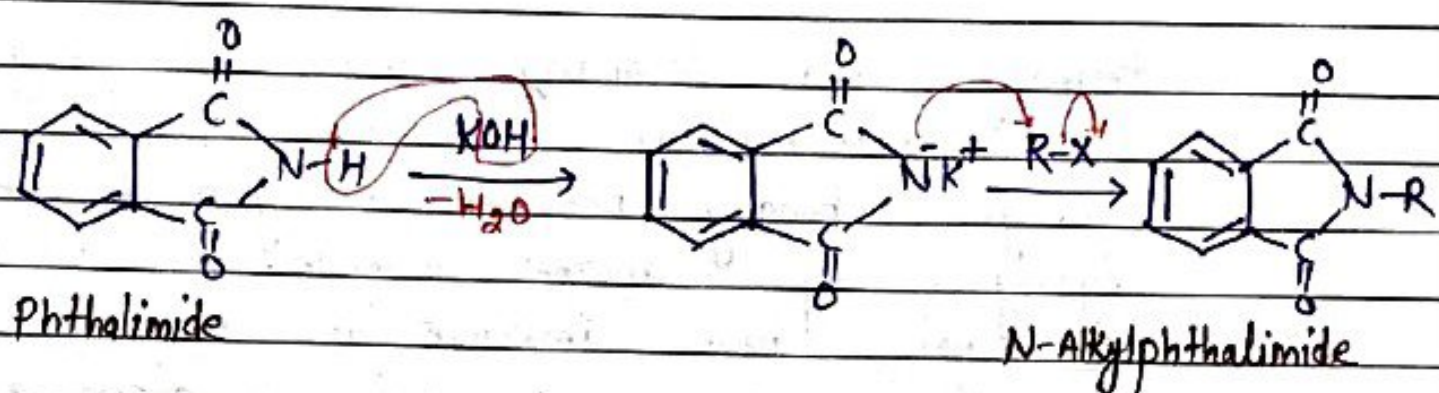


4. Reduction of Amides:-



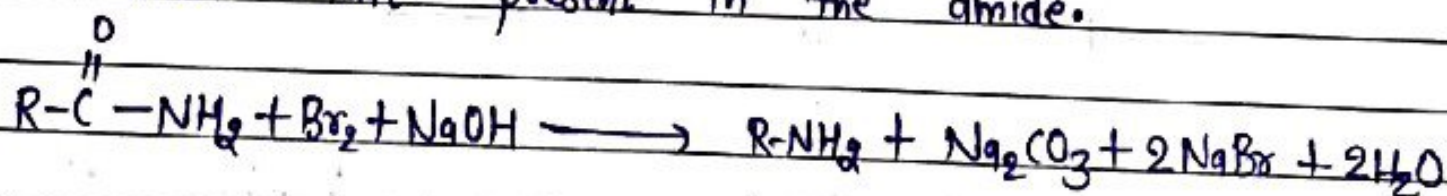
★

5. Gabriel Phthalimide synthesis:- Gabriel Phthalimide synthesis is used for the preparation of primary amines.



6. Hoffmann Bromamide Degradation Reaction:-
Reagent $\rightarrow X_2 + NaOH/KOH$

This method is also used for preparation of 1° Amines. The amine so formed contains one carbon less than that present in the amide.



PHYSICAL PROPERTIES:-

Lower aliphatic amines are soluble in water because they can form hydrogen bonds with water molecules.

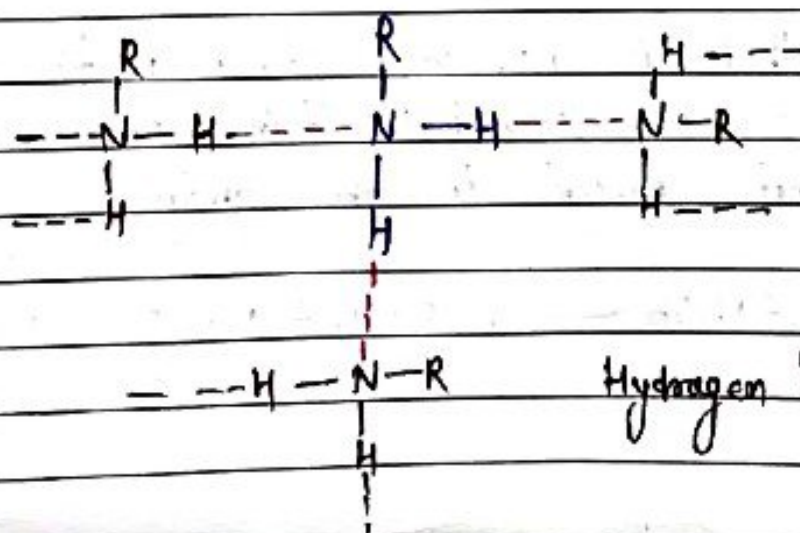
Solubility decreases with increase in molar mass of amines due to increase in size of the hydrophobic alkyl part.

Primary and secondary amines are engaged in intermolecular association due to hydrogen bonding between nitrogen of one and hydrogen of another molecule. Tertiary amines do not have intermolecular association due to the absence of hydrogen atom available for hydrogen bond formation.

Hydrogen bonding $\rightarrow 1^\circ \text{Amine} > 2^\circ \text{Amine} > 3^\circ \text{Amine}$
(Zero)

Therefore, the order of boiling points of isomeric amines -

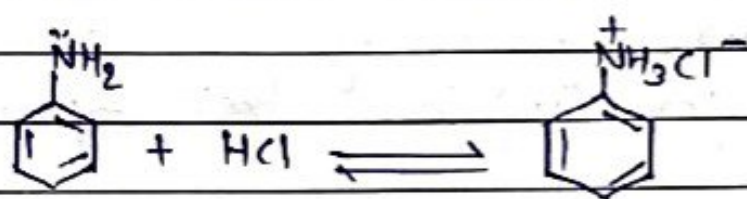
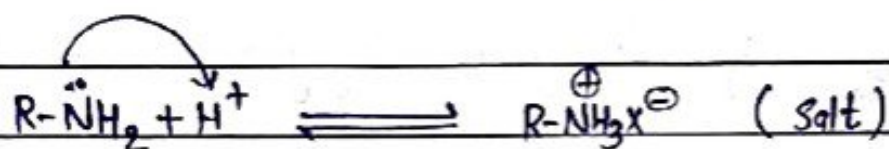
Primary $>$ Secondary $>$ Tertiary



Hydrogen bonding in 1° Amine

CHEMICAL PROPERTIES

1. Basic Nature of Amines:- Amines react with acids to form salts.



★ Longer the value of K_b or smaller the value of pK_b , stronger is the base.

$$\text{Basicity} \propto K_b \propto \frac{1}{pK_b}$$

→ Aliphatic amines are stronger bases than ammonia due to +I effect of alkyl groups leading to high electron density on the nitrogen atom.

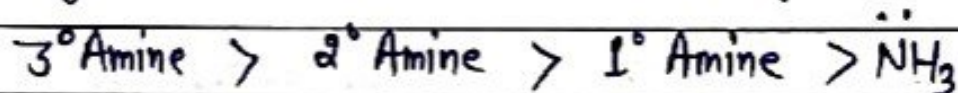
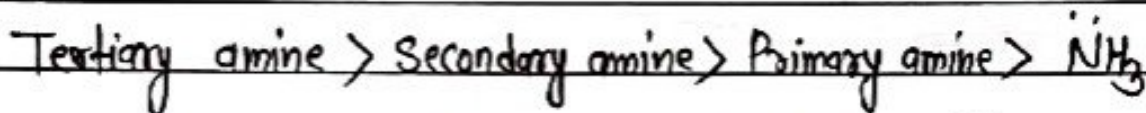
→ Aromatic amines are weaker bases than ammonia due to the electron withdrawing nature of the aryl group.

Alkylamines (Alkanamines) versus ammonia:-



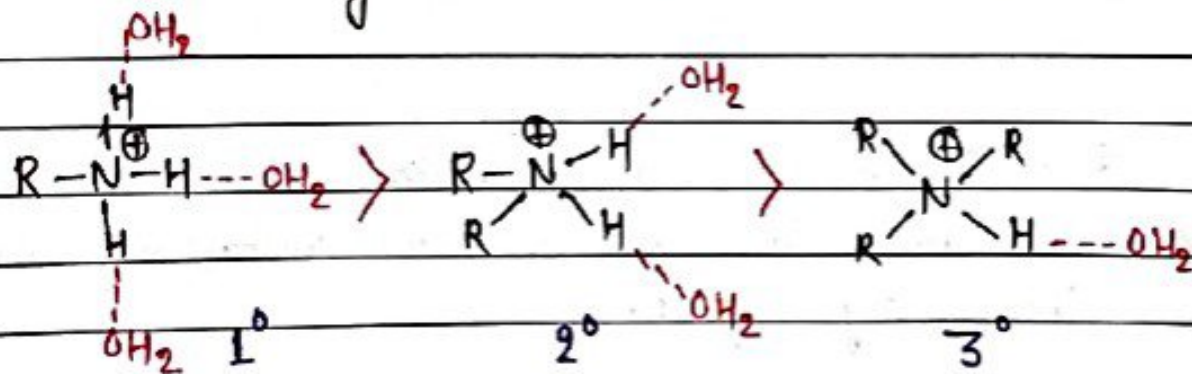
Due to +I effect of alkyl group, conjugate cation stabilize.

So the order of basicity of amines in the gaseous phase follows the expected order :-

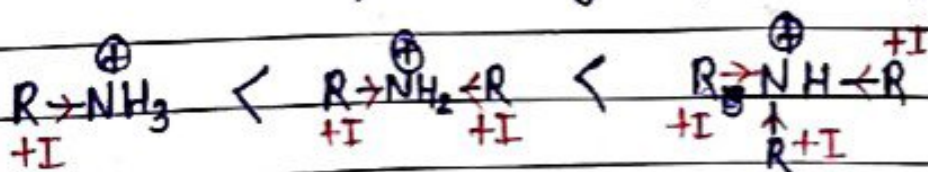


The trend is not regular in the aqueous phase -

★ In the aqueous phase, the substituted ammonia cations get stabilized not only by electron releasing effect of the alkyl group (+I) but also by solvation with water molecules.



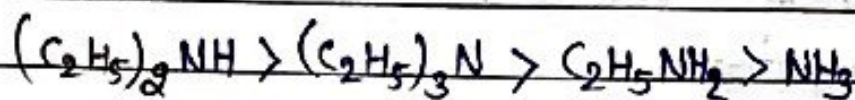
ORDER of stability of ions by solvation



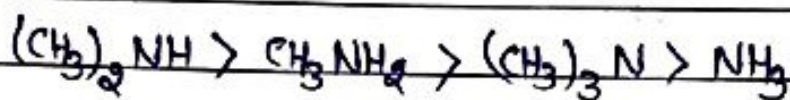
ORDER of stability of ions by electronic effect (+I)

ORDER of basicity in aqueous solution

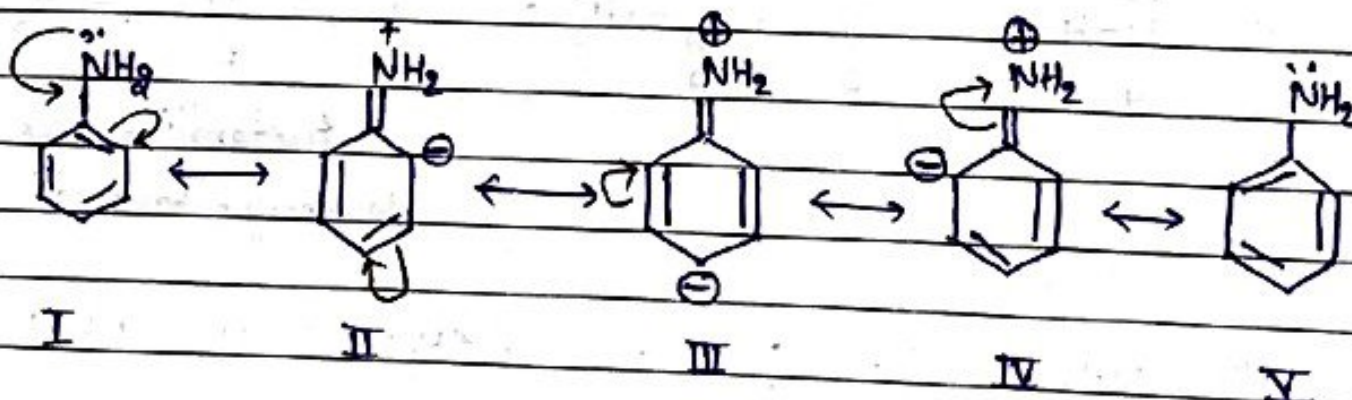
If $R = -C_2H_5$



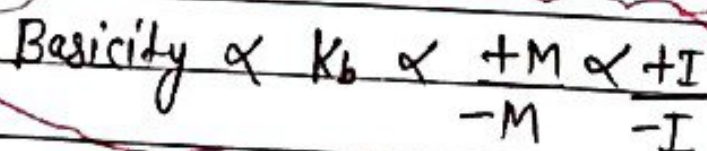
If $R = -CH_3$



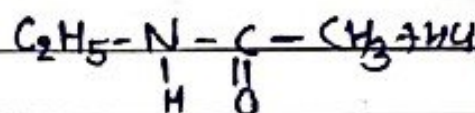
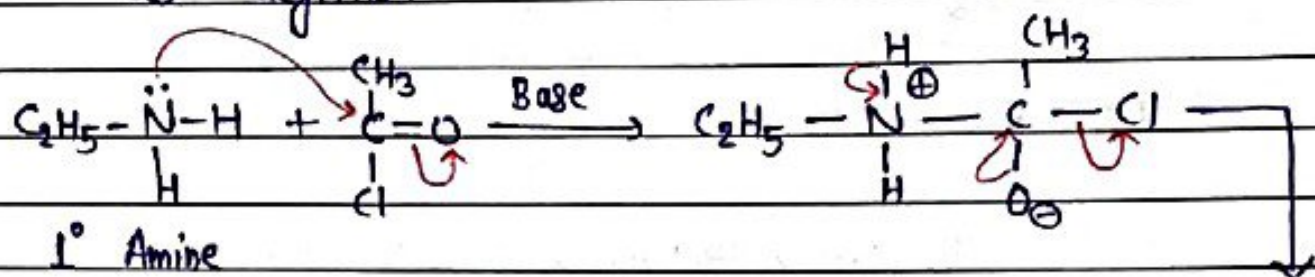
(B) Arylamines versus ammonia:- pK_b value of aniline is quite high (K_b is low). It is because in aniline or other arylamines, the $-NH_2$ group is attached directly to the benzene ring. It results in the unshared electron pair on nitrogen atom to be in conjugation with the benzene ring and thus making it less available for protonation.



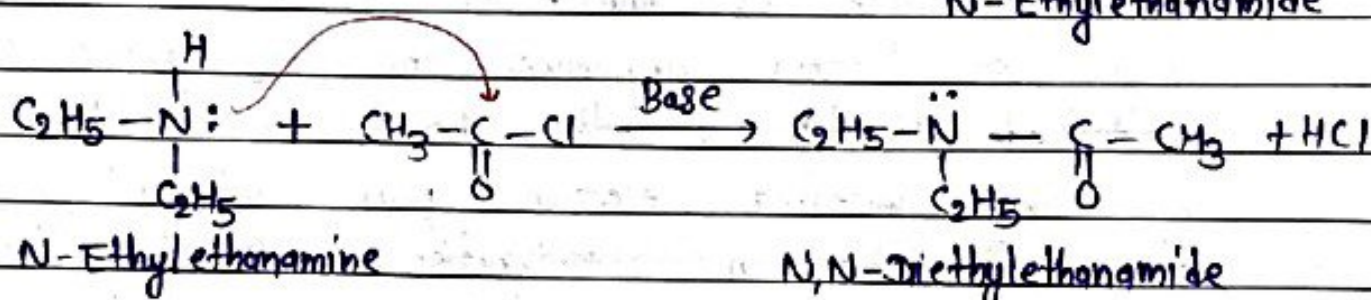
ORDER of basicity:- Aliphatic amines > Ammonia > Aniline



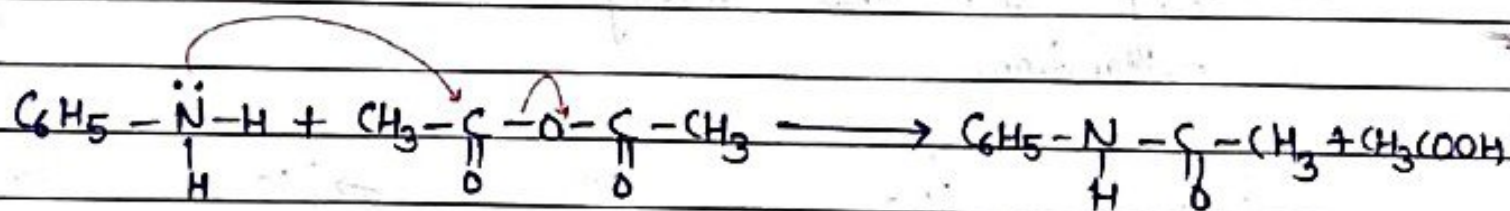
ACYLATION:- Aliphatic & Aromatic primary and secondary amines react with acid chlorides, anhydrides and esters by nucleophilic substitution reaction. This reaction is known as acylation.



N-Ethylethanamide

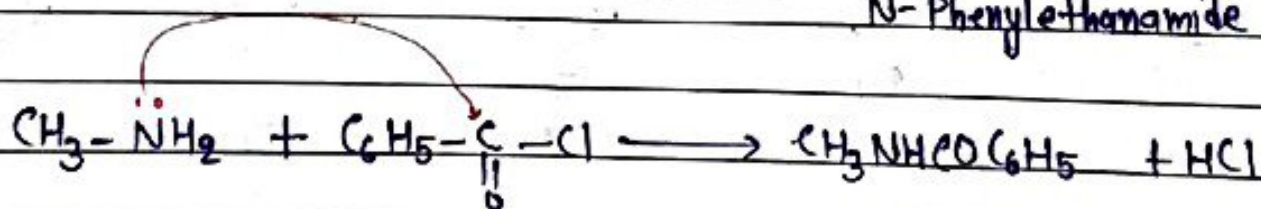


N,N-Diethylethanamide



Acetanilide or

N-Phenylethanamide



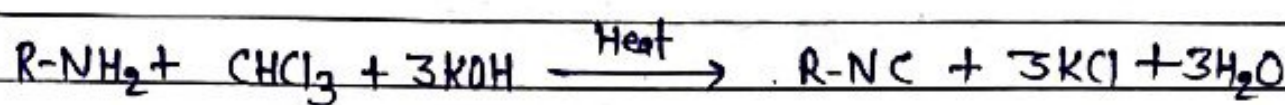
Methanamine

Benzoyl chloride

N-Methylbenzamide

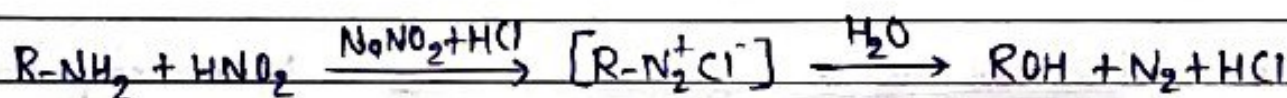
→ Test for 1° Amines (Aliphatic & Aromatic)

CARBYLAMINE REACTION:- Aliphatic and aromatic primary amines on heating with chloroform and ethanolic potassium hydroxide form isocyanide or carbylamines which are foul smelling substances.
Secondary and tertiary amines do not show this reaction.

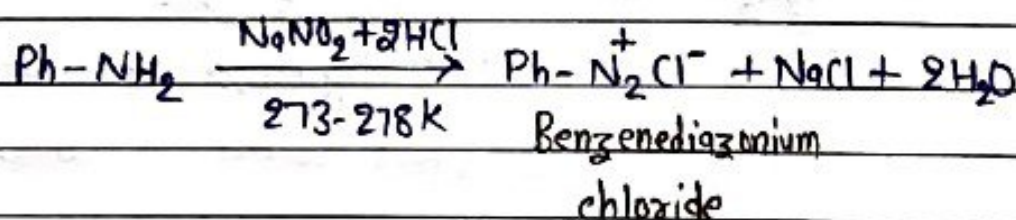


REACTION WITH NITROUS ACID:-

(A) Primary Aliphatic Amines:-



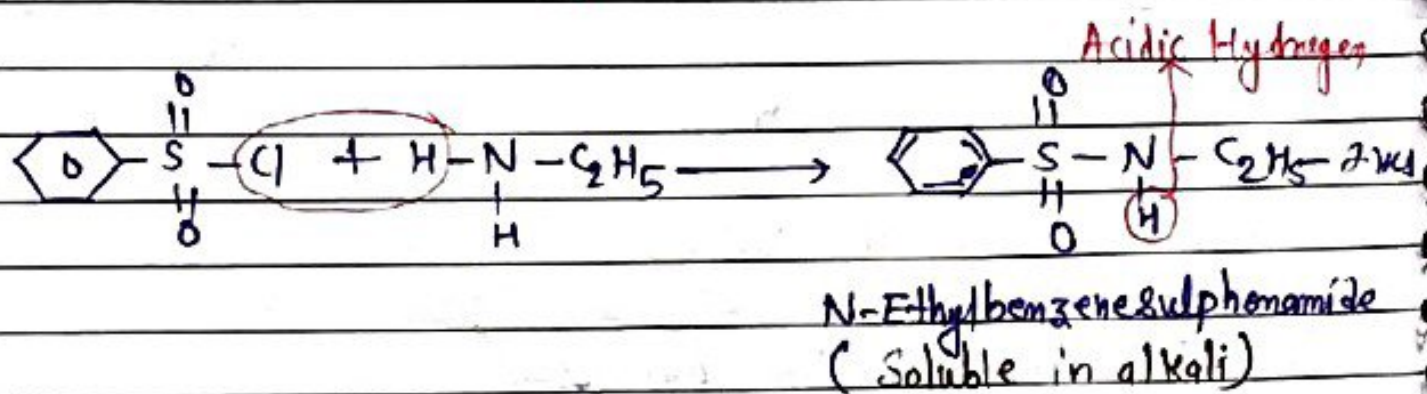
(B) Aromatic Amines:-



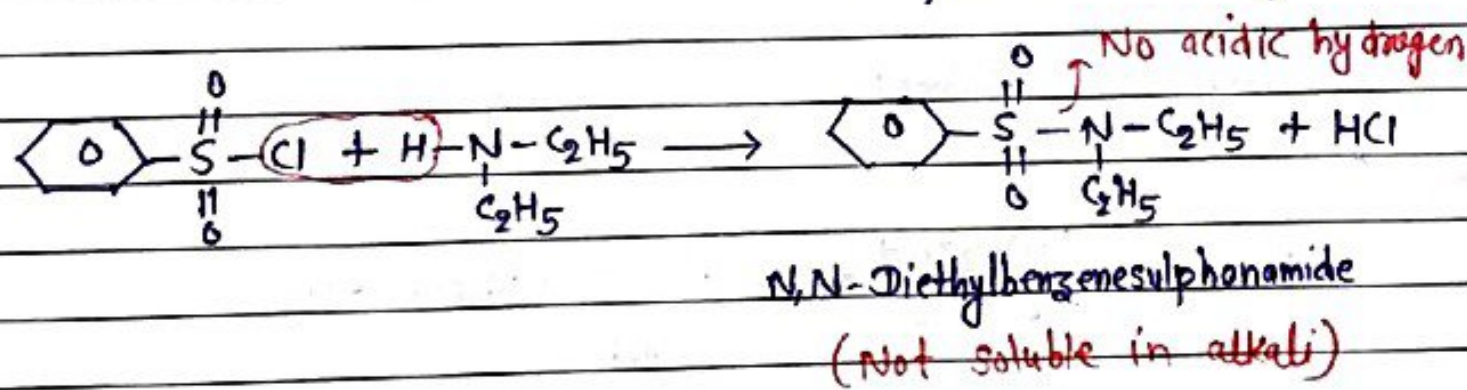
REACTION WITH ARYLSULPHONYL CHLORIDE:-

Benzenesulphonyl chloride ($C_6H_5SO_2Cl$), which is also known as Hinsberg's reagent, reacts with primary and secondary amines to form sulphonamides.

(a) Reaction with primary amines:-



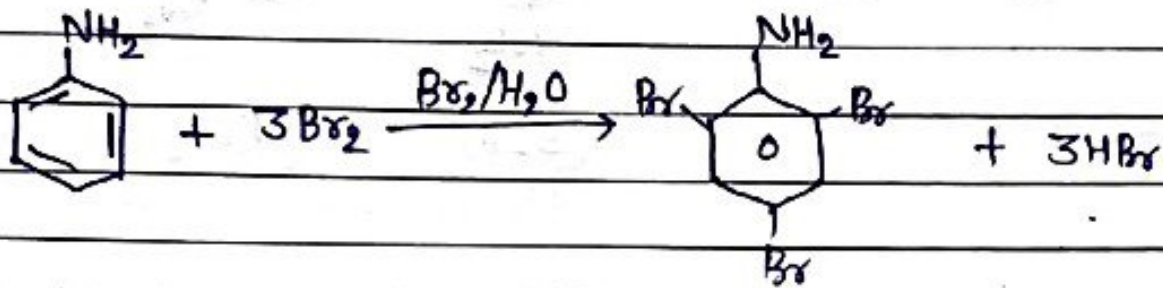
(b) Reaction with secondary amines:- N,N-diethylbenzenesulphonamide is formed.



(c) Tertiary amines:- Tertiary amines do not react with benzenesulphonyl chloride.

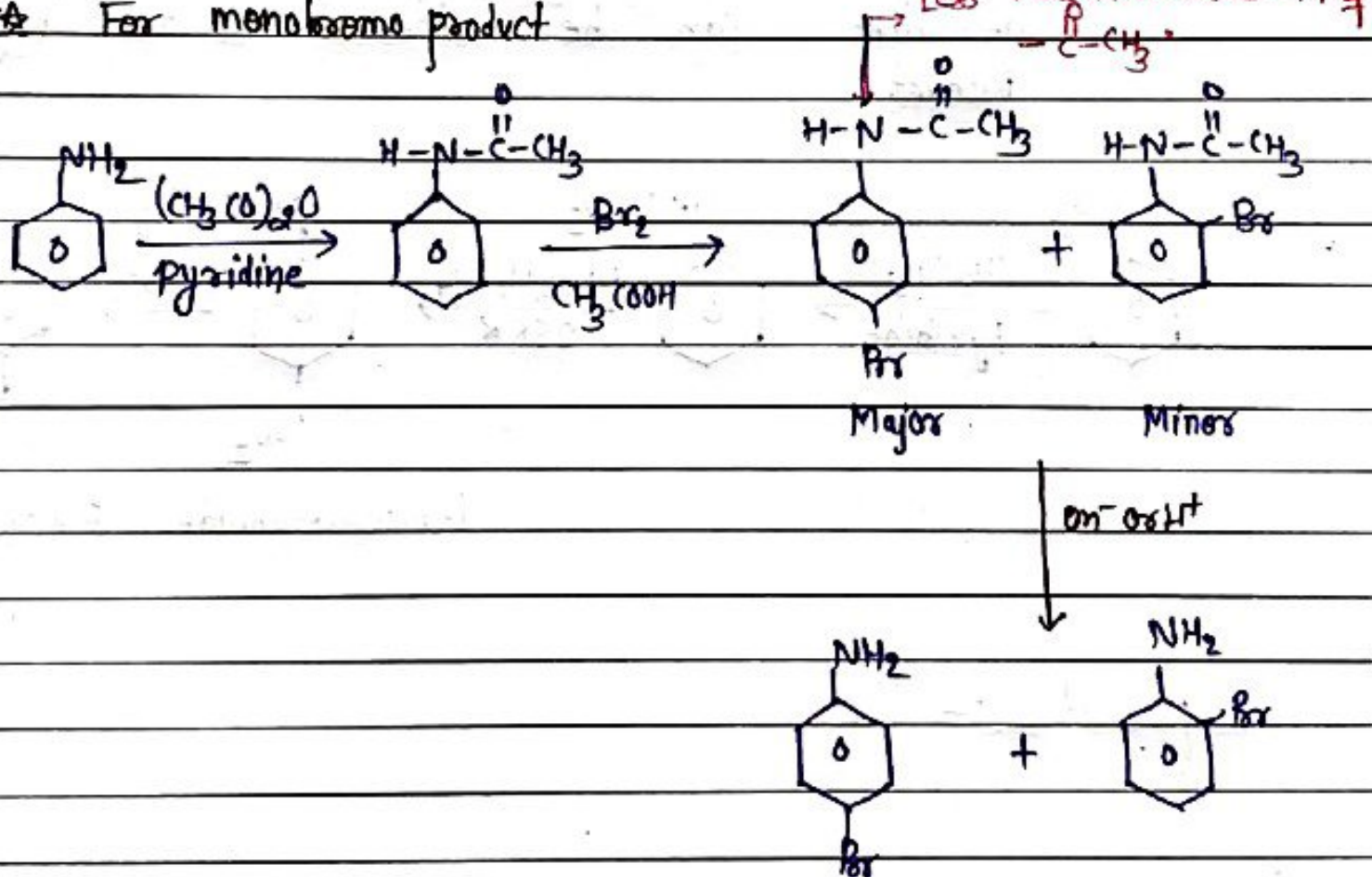
ELECTROPHILIC SUBSTITUTION:- $-NH_2$ group is ortho and para directing and a powerful activating group.

(A) Bromination:- Aniline reacts with bromine water at room temperature to give a white precipitate of 2,4,6-tribromoaniline.



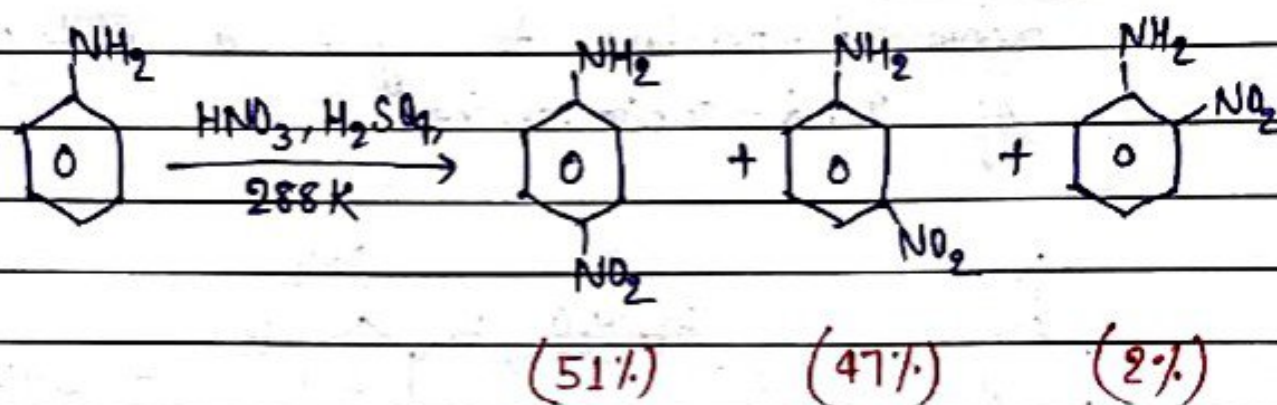
2,4,6-Tribromoaniline

★ For monobromo product

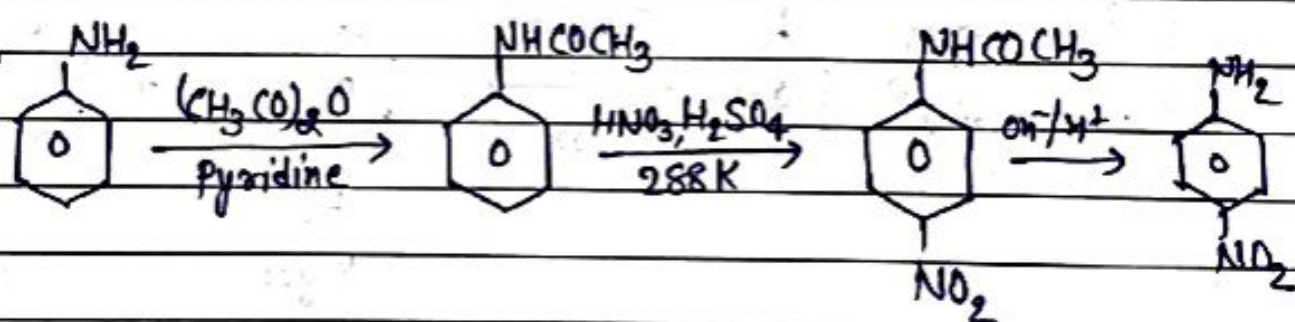


4-Bromoaniline

B) Nitration:- In the strongly acidic medium, aniline is protonated to form the anilinium ion which is meta directing.

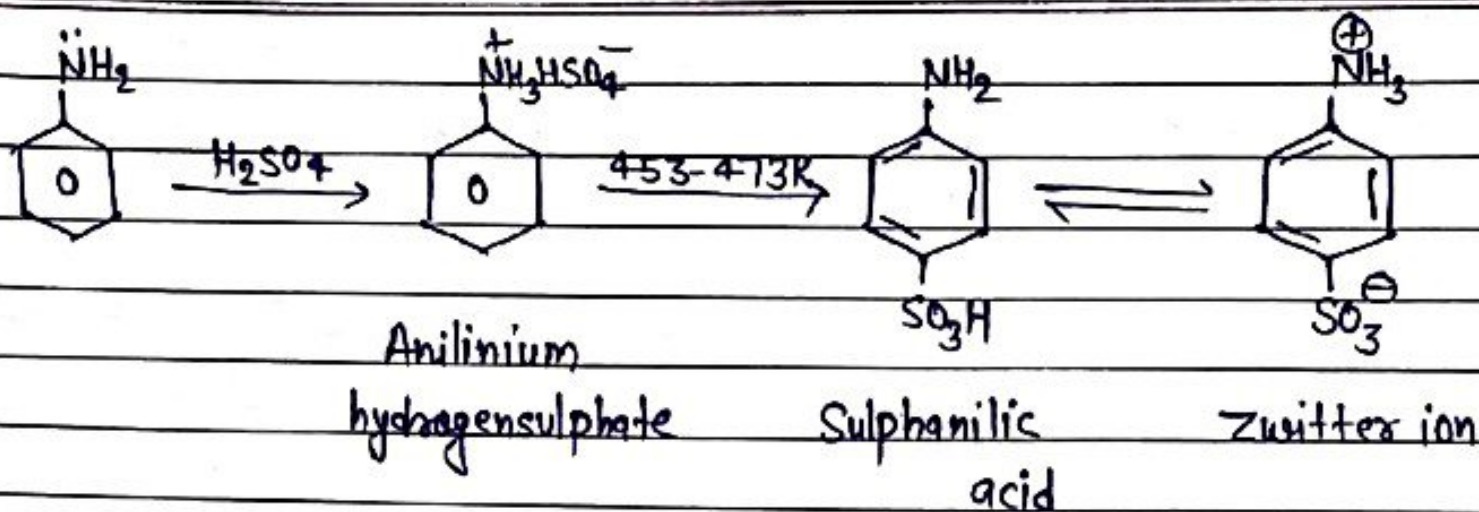


★ By protecting the $-\text{NH}_2$ group by acetylation reaction with acetic anhydride, the nitration reaction can be controlled and the p-nitro derivative can be obtained as the major product.



P-nitroacetanilide
P-nitroaniline

Sulphonation:-



→ Aniline does not undergo Friedel-Crafts reaction (alkylation and acetylation) due to salt formation with aluminium chloride, Lewis acid, which is used as a catalyst. Due to this, nitrogen of aniline acquires positive charge and hence acts as a strong deactivating group.

THE
END